NIKHIL GOVINDARAJU

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EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA **Jan 2024 - Dec 2025**
Master of Science in Computer Science | GPA: 3.81/4.0 **Los Angeles, CA**
Coursework: Algorithms, Database Systems, Web Technologies, Information Retrieval, Machine Learning for Data Science
Leadership: Viterbi Graduate Mentor - mentoring graduate students on coursework planning, internships, and navigating USC resources

JSS SCIENCE AND TECHNOLOGY UNIVERSITY **Aug 2017 - Jul 2021**
Bachelor of Engineering in Information Science and Engineering | GPA: 9.01/10.0 **Mysore, India**
Coursework: Data Structures, Object Oriented Design, Data Mining, Operating Systems, Distributed Computing, Computer Networks, Software Engineering

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
Backend & Systems: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, Spring Boot
Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap
Databases: PostgreSQL, MongoDB, MySQL, DynamoDB, Prisma ORM
Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch, RDS), GCP, Docker, Kubernetes, GitHub Actions, CI/CD
Testing & Tools: Pytest, Playwright, Cypress, Jira, Postman, Agile/Scrum
AI & LLM Systems: LangChain, LangGraph, RAG Pipelines, OpenAI API, HuggingFace, Prompt Engineering

EXPERIENCE

Jubilant HollisterStier LLC **May 2025 - Aug 2025**
Software Engineering Intern **Spokane, WA**

- Architected and deployed scalable backend services (Express.js + PostgreSQL) on AWS (EC2, RDS), reducing reporting latency by 50% in an FDA-regulated healthcare environment
- Designed normalized relational schemas and built high-performance REST APIs with server-side caching for scalable scheduling workflows
- Implemented CI/CD pipelines (GitHub Actions) with automated testing, reducing rollback incidents by 20%
- Introduced feature flags and staged deployments, enabling zero-downtime releases in an FDA-regulated production environment
- Built React + Recharts analytics dashboards consuming internal REST APIs, delivering real-time technician utilization and shift planning insights to operations managers

Exinous Technologies India Pvt Ltd **Sep 2022 - Nov 2023**
Software Development Engineer **Mysore, India**

- Improved application performance through API batching, image compression, and code-splitting, reducing load times by 25%
- Improved client-side rendering stability and layout consistency, reducing cumulative layout shift by 10%
- Developed inventory management modules for an ERP system using Vue.js + TypeScript, building reusable components against backend service APIs and reducing feature rework by 20%
- Wrote Cypress E2E and API-level tests for critical application flows, automating regression coverage and reducing manual QA effort by 25%

Bosch Global Software Technologies **Jun 2021 - Sep 2022**
Associate Software Engineer **Bangalore, India**

- Improved embedded and signal-processing features by tuning radar classification logic and optimizing UI redraw paths, reducing false detections by 25% and improving UI frame rates by 30%, working in an Agile sprint-based development environment.
- Owned migration of 15K+ lines of legacy Python 2 systems to Python 3, improving runtime stability and long-term maintainability in a production
- Implemented firmware update and logging workflows, contributing to multiple Bosch Measure 200 releases with zero critical production issues.

PROJECTS

LeaveNow | Next.js, NestJS, Spring Boot, PostgreSQL, Redis, Docker **Link**

- Architected a polyglot microservices backend (Spring Boot + NestJS) for multimodal ETA planning across drive, transit, walk, and cab modes, designed service boundaries, inter-service communication, and API contracts from scratch
- Engineered a fault-tolerant ETA computation engine using Resilience4j circuit breakers, handling partial upstream API failures while preserving real-time routing responsiveness
- Applied Redis caching with traffic, weather, and buffer rule evaluation, cutting routing latency by 35% through optimized API orchestration
- Containerized all services with Docker, configured CI/CD pipelines, and wrote integration tests that reduced cross-service defects by 30%

Realtime Weather Dashboard Application | Flask, MEAN, Swift, GCP **Link**

- Built a Node.js + Flask backend serving real-time geolocation weather data to Angular and iOS clients, integrating Tomorrow.io and Google Maps APIs with sub-150ms response times
- Implemented MongoDB-backed caching to deduplicate external API calls, reducing redundant requests by 30%
- Deployed backend services on GCP with autoscaling and health monitoring, achieving 99.9% uptime under variable load conditions

Secure EHR AI | React, FastAPI, PostgreSQL, AWS **Link**

- Designed a FastAPI backend with role-based access control (RBAC) enforcing strict permission boundaries across doctor, patient, and auditor roles
- Implemented encrypted audit logging with RSA/AES and SHA-256 hash-chaining, providing HIPAA-compliant tamper-evident record trails with 100% modification detection
- Architected a LangChain RAG pipeline integrated into the backend service layer, improving clinical information retrieval speed by 50%

CERTIFICATIONS

AWS Cloud Practitioner | Agentic AI for Developers | OCI AI Foundations | Google AI Essentials | ML & AI by NASSCOM | Certified Scrum Master